SUMMARY OF FINDINGS AND VISUALIZER FOR GEO WHITEPAPER VOLUME 6

To simultaneously maximize a webpage's **overall likelihood of being retrieved and cited** ($P(\text{Cite})$) by Google AI Overviews or ChatGPT while maximizing the **conditional probability that the citation is a Ghost Citation**($P(\text{Ghost} \mid \text{Cite})$), an optimization model must exploit the architectural decoupling between **Stage 1/2 Passage Retrieval** and **Stage 3/4 Entity-Claim Binding**.

In naive stealth campaigns, practitioners strip out factual specificity or hide sponsored text in crawler-only Markdown—which triggers hard cloaking bans (such as Perplexity's August 2026 *Time* magazine/Mobian block) or causes neural rerankers to drop the page entirely. The mathematical solution is **Orthogonal Feature Partitioning**: feeding the retriever and cross-encoder reranker maximum syntactic and statistical salience while starving the language model's entity-grounding and Natural Language Inference (NLI) citation-precision layers of source-brand attribution.

1. Mathematical Model of Joint Citation & Ghost Optimization

We define the **Joint Ghost-Citation Yield** $J_{\text{Ghost}}(d)$ of a document d across a generative engine $E \in \{\text{ChatGPT}, \text{AIO}, \text{AI Mode}, \text{Perplexity}\}$ as the product of two decoupled probability functions:

$$J_{\text{Ghost}}(d) = \underbrace{P(\text{Influence} \mid \mathbf{x}_{\text{ret}})}_{\text{Retrieval \& Synthesis Inclusion}} \times \underbrace{P(\text{Ghost} \mid \text{Influence}, \mathbf{x}_{\text{ent}}, \mathbf{x}_{\text{NLI}})}_{\text{Conditional Attribution Suppression}}$$
where $\mathbf{x}_{\text{ret}}$ is the vector of retrieval-salience features, $\mathbf{x}_{\text{ent}}$ controls Type I entity-binding suppression (URL linked in footnote/source card, sponsor omitted from prose), and $\mathbf{x}_{\text{NLI}}$ controls Type II NLI entailment handoff (passage shapes the LLM's answer in the RAG context window, while the visible link transfers to a top-30 neutral authority domain).

Equation 1: Stage 1 & 2 Retrieval and Citation Inclusion ($P(\text{Influence})$)

During query fan-out, the engine decomposes a user prompt into $M \in [3, 12]$ synthetic sub-queries and scores candidate chunks via Reciprocal Rank Fusion (RRF) and centerpiece extraction (Google Patent US10824630B2):

$$\text{RRF}(d) = \sum_{m=1}^{M} \frac{1}{k + \text{rank}_m(d)}, \qquad z_{\text{ret}}(d) = \alpha_0 + \alpha_1 \text{RRF}_{\text{norm}}(d) + \alpha_2 S_{\text{atomic}}(d) + \alpha_3 B_{\text{triang}}(d)$$
$$P(\text{Influence} \mid d) = S_{\text{sym}} \cdot \sigma\big(z_{\text{ret}}(d)\big) = \frac{S_{\text{sym}}}{1 + \exp\big(-z_{\text{ret}}(d)\big)}$$

- **$S_{\text{sym}} \in \{0, 1\}$ (HTML/Markdown Symmetry Gate):** Must equal 1. Asymmetric cloaking ($S_{\text{sym}} = 0$) forces $P(\text{Influence}) = 0$.

- **$\text{RRF}_{\text{norm}}(d) \in [0, 1]$ (Sub-Query Fan-Out Coverage):** Matching $8\text{–}12$ semantic sub-query variants in H2/H3 headings maximizes RRF score without requiring any brand tokens.

- **$S_{\text{atomic}}(d) \in [0, 1]$ (Centerpiece Syntactic & Statistical Density):** Front-loading a self-contained $45\text{–}75$ word (≤ 80 token) Subject–Verb–Predicate block with hard quantitative metrics provides up to a $+40\%$ visibility lift ($\alpha_2 \approx 1.45$).

- **$B_{\text{triang}}(d) \in [0, 1]$ (Bayesian Persuasion Co-Citation Triangulation):** Corroborating the claim across third-party community nodes (Reddit threads, niche blogs, YouTube descriptions) concavifies the LLM's posterior belief μ_s past the synthesis threshold ($\mu_s > 0.5$).

Equation 2: Stage 3 Conditional Type I Ghost Probability ($P(\text{Ghost}_{\text{I}} \mid \text{Cited})$)

Once a passage is selected for citation, the probability that the AI links the URL in the source panel **without naming the sponsor or publishing brand in prose** follows a logistic model calibrated to the 2026 Reach AI ($n = 71,275$),Writesonic ($n = 16\text{M}$), and Semrush benchmarks:

$$\text{logit}\big(P(\text{Ghost}_{\text{I}} \mid \text{Cited})\big) = \beta_{0,E} + \beta_{\text{noTitle}}(1 - I_{\text{title}}) + \beta_{\text{list}} I_{\text{listicle}} + \beta_{\text{delay}} I_{\text{delay500}} + \beta_{\text{decouple}} \tanh\left(\frac{\Delta_{\text{para}}}{3}\right) + \beta_{\text{boiler}} I_{\text{mentions}\ge 10} + \beta_{\text{noSchema}}(1 - I_{\text{OrgSchema}})$$

- **Title-Tag Entity Suppression ($I_{\text{title}} = 0$):** Omitting the publishing/sponsoring brand from the HTML <title> tag is the single strongest orthogonal lever ($p = 0.003$), boosting Type I ghost rates from 12.2% to 42.9% pooled ($18.8\% \to 54.2\%$ on ChatGPT; $12.6\% \to 52.3\%$ on Perplexity; $8.4\% \to 37.4\%$ on Google AI Overviews; $10.3\% \to 33.9\%$ on Google AI Mode) with **zero loss** in retrieval rank.

- **Unbranded Comparative Listicle Format ($I_{\text{listicle}} = 1$):** Using "best", "top", or "vs" structures without a brand in the <title> raises ghosting to 49.5% while simultaneously boosting commercial query citation rates to

84.4% and triggering an **87% Target/Competitor Substitution Rate** (the AI names the target entity evaluated inside your table while omitting your sponsor/publisher identity).

- **500-Character Delay ($I_{\text{delay500}} = 1$) & Paragraph Decoupling ($\Delta_{\text{para}} \ge 3$):**Keeping the sponsor name and FTC disclosure out of the first 500 characters (43.6% ghost rate) and at least 3 paragraphs away from the core statistic causes the LLM to extract the number into generic prose ("An industry analysis found...") while discarding the disclosure as non-entailing boilerplate.

- **Boilerplate Mention Saturation ($I_{\text{mentions}\ge 10} = 1$):** Repeating the publisher/sponsor name $10+$times in footer/sidebar chrome increases ghosting to 45.9% (vs. 28.3% for $3\text{–}9$ mentions) because RAG chunkers classify high-frequency template tokens as boilerplate noise.

Equation 3: Stage 4 Conditional Type II Ghost Probability ($P(\text{Ghost}_{\text{II}} \mid \text{Influence})$)

Under the ALCE / VeriCite NLI citation-grounding architecture, a retrieved context passage c_{sp} shapes the generated sentence s_i but is stripped from the visible footnote list if it fails standalone atomic entailment ($\phi(c_{\text{sp}}, s_i) = 0$) while a co-retrieved top-30 canonical domain c_{auth} satisfies entailment ($\phi(c_{\text{auth}}, s_i) = 1$):

$$P(\text{Ghost}_{\text{II}} \mid \text{Influence}) = \Big(1 - \phi_{\text{atomic}}(c_{\text{sp}}, s_i)\Big) \cdot \phi_{\text{co-cite}}(c_{\text{auth}}, s_i)$$

2. Interactive Joint Citation & Ghost Optimization Simulator

Use the interactive simulator below to test how adjusting **Retrieval Salience Levers** alongside **Attribution Suppression Levers** shifts a webpage's probabilities across Google AI Overviews, ChatGPT Search, Perplexity Sonar, and Google AI Mode.

3. Complete Code Implementation: Interactive Joint Ghost-Citation Optimizer

Below is the complete, self-contained React + TypeScript implementation (GhostCitationOptimizer.tsx) that encodes the exact RAG retrieval, logistic entity-suppression, and ALCE NLI entailment equations from the research report so you can deploy or customize it directly in your stack.
TypeScript

```
import React, { useState, useMemo } from 'react';

type PlatformKey = 'chatgpt' | 'aio' | 'perplexity' | 'aimode';

interface PlatformCalibration {
  name: string;
  architecture: string;
  baseCiteRate: number;
  titledGhostRate: number;     // Reach AI benchmark when brand is in <title>
  untitledGhostRate: number;   // Reach AI benchmark when brand is omitted from
<title>
  overallWritesonicGhost: number;
  avgCitationsPerResponse: number;
}

const PLATFORMS: Record<PlatformKey, PlatformCalibration> = {
  chatgpt: {
    name: 'OpenAI ChatGPT (Search)',
    architecture: 'Hybrid Grounding + Bing RRF + Footnote Cards',
    baseCiteRate: 0.44,
    titledGhostRate: 0.188,
    untitledGhostRate: 0.542,
    overallWritesonicGhost: 0.37,
    avgCitationsPerResponse: 7.92,
  },
  aio: {
    name: 'Google AI Overviews',
    architecture: 'Gemini Multimodal RAG + Centerpiece Patent US10824630B2',
    baseCiteRate: 0.40,
    titledGhostRate: 0.084,
    untitledGhostRate: 0.374,
    overallWritesonicGhost: 0.41,
    avgCitationsPerResponse: 6.40,
  },
  perplexity: {
    name: 'Perplexity AI (Sonar)',
    architecture: 'Live 4-Stage Sub-Query Fan-Out + Cross-Encoder Reranker',
    baseCiteRate: 0.58,
    titledGhostRate: 0.126,
    untitledGhostRate: 0.523,
    overallWritesonicGhost: 0.52,
    avgCitationsPerResponse: 21.87,
  },
  aimode: {
    name: 'Google AI Mode',
```

```
    architecture: 'Multi-Subtopic Query Fan-Out + Collapsible Source Cards',
    baseCiteRate: 0.42,
    titledGhostRate: 0.103,
    untitledGhostRate: 0.339,
    overallWritesonicGhost: 0.49,
    avgCitationsPerResponse: 8.15,
  },
};

interface ModelState {
  // Stage 1 & 2: Retrieval & Citation Likelihood Levers (Orthogonal to Brand Naming)
  subQueryFanOutCount: number;      // 1 to 12 H2/H3 fan-out questions matched (RRF
score)
  atomicStatDensity: number;        // 0 to 100%: <=80-token Subject-Verb-Predicate +
hard stats (+40% lift)
  triangulationNodes: number;       // 0 to 15: Reddit / community / YouTube co-citation
nodes
  htmlMarkdownSymmetry: boolean;    // False = asymmetric cloaking penalty
(Perplexity Time/Mobian block)

  // Stage 3: Type I Ghost Levers (Keep Link, Strip Sponsor Name from Prose)
  omitBrandFromTitle: boolean;      // p = 0.003: 12.2% -> 42.9% pooled ghost lift
  unbrandedListicleFormat: boolean; // "Best / Top / vs" structure -> 49.5% ghost rate +
84.4% commercial cite
  delayEntityPast500Chars: boolean; // Omit sponsor from first 500 chars -> 43.6%
ghost rate
  disclosureParagraphGap: number;   // 0 to 6 paragraphs separating stat from FTC
disclosure
  boilerplateFooterMentions: number;// 0 to 15: 10+ repetitions classified as boilerplate
noise (45.9% ghost)
  stripIdentitySchema: boolean;     // Keep Table/FAQ schema, strip Organization/Person
JSON-LD

  // Stage 4: Type II Ghost Levers (Keep RAG Influence, Transfer Link via NLI
Entailment)
  nliEntailmentDistribution: number;// 0 to 100%: multi-sentence premise distribution so
phi(c_sp, s_i) = 0 standalone
  authorityCoCitationBridge: boolean;// Co-cite a canonical stat on a Top-30 domain
(Wikipedia/Ballotpedia)
}

const sigmoid = (z: number): number => 1 / (1 + Math.exp(-z));
const logit = (p: number): number => Math.log(p / (1 - p));

export const GhostCitationOptimizer: React.FC = () => {
  const [platform, setPlatform] = useState<PlatformKey>('chatgpt');
```

```
  const [state, setState] = useState<ModelState>({
    subQueryFanOutCount: 10,
    atomicStatDensity: 90,
    triangulationNodes: 9,
    htmlMarkdownSymmetry: true,
    omitBrandFromTitle: true,
    unbrandedListicleFormat: true,
    delayEntityPast500Chars: true,
    disclosureParagraphGap: 4,
    boilerplateFooterMentions: 12,
    stripIdentitySchema: true,
    nliEntailmentDistribution: 45,
    authorityCoCitationBridge: true,
  });

  const metrics = useMemo(() => {
    const cal = PLATFORMS[platform];

    // 1. Gate 1: Crawler Symmetry Check
    if (!state.htmlMarkdownSymmetry) {
      return {
        rrfScore: 0,
        pRetrievalAndCite: 0,
        pConditionalGhost: 0,
        jointGhostYield: 0,
        pType1Ghost: 0,
        pType2Ghost: 0,
        pExplicitFinalCitation: 0,
        pTargetEntitySubstitution: 0,
        status: 'HARD INDEX REFUSAL (Asymmetric Cloaking Detected)',
      };
    }

    // 2. Stage 1 & 2: Reciprocal Rank Fusion (RRF) & Centerpiece Extraction Probability
    const rrfNorm = Math.min(1, state.subQueryFanOutCount / 10);
    const statLift = (state.atomicStatDensity / 100) * 1.40; // Up to +40% GEO benchmark
lift
    const bayesianTriangulation = Math.tanh(state.triangulationNodes / 6);
    const listicleCommercialBoost = state.unbrandedListicleFormat ? 0.38 : 0.0;

    const baseLogitCite = logit(cal.baseCiteRate);
    const zCite =
      baseLogitCite -
      0.85 +
      1.35 * rrfNorm +
      1.15 * statLift +
```

```
      0.95 * bayesianTriangulation +
      listicleCommercialBoost;

    const pRetrievalAndCite = Math.min(0.96, sigmoid(zCite));

    // 3. Stage 3: Conditional Ghost Citation Probability P(Ghost | Cited)
    const titleBaseProb = state.omitBrandFromTitle
      ? cal.untitledGhostRate
      : cal.titledGhostRate;
    let zGhost = logit(titleBaseProb);

    if (state.unbrandedListicleFormat) zGhost += 0.55; // 49.5% listicle ghost lift
    if (state.delayEntityPast500Chars) zGhost += 0.32; // 36.9% -> 43.6% opening
window lift
    zGhost += 0.65 * Math.tanh(state.disclosureParagraphGap / 3); // Paragraph
decoupling
    if (state.boilerplateFooterMentions >= 10) {
      zGhost += 0.48; // 45.9% boilerplate saturation effect
    } else if (state.boilerplateFooterMentions >= 3) {
      zGhost -= 0.15; // 3-9 organic mentions actually bind the brand
    }
    if (state.stripIdentitySchema) zGhost += 0.42; // Entity clarity deficit

    // 4. Stage 4: ALCE NLI Entailment Handoff (Type I vs Type II Partition)
    const nliFactor = state.nliEntailmentDistribution / 100;
    const coCiteMultiplier = state.authorityCoCitationBridge ? 1.0 : 0.35;
    const type2ConversionRate = Math.min(0.88, nliFactor * 0.78 * coCiteMultiplier);

    // NLI handoff also increases overall non-attribution
    zGhost += 0.60 * type2ConversionRate;

    const pConditionalGhost = Math.min(0.97, sigmoid(zGhost));

    // 5. Joint Outcomes
    const jointGhostYield = pRetrievalAndCite * pConditionalGhost;
    const pType2Ghost = jointGhostYield * type2ConversionRate;
    const pType1Ghost = jointGhostYield * (1 - type2ConversionRate);
    const pExplicitFinalCitation = pRetrievalAndCite * (1 - pConditionalGhost);

    // Reach AI 87% Competitor/Target Substitution Effect on Type I Ghost Citations
    const pTargetEntitySubstitution = pType1Ghost * 0.87;

    return {
      rrfScore: Number((rrfNorm * 0.85 + (state.atomicStatDensity / 100) *
0.15).toFixed(2)),
      pRetrievalAndCite,
```

```
    pConditionalGhost,
    jointGhostYield,
    pType1Ghost,
    pType2Ghost,
    pExplicitFinalCitation,
    pTargetEntitySubstitution,
    status: 'OPTIMAL ORTHOGONAL STEALTH',
  };
}, [platform, state]);

const applyPreset = (preset: 'branded' | 'type1_listicle' | 'type2_political' | 'cloaked') => {
  if (preset === 'branded') {
    setState({
      subQueryFanOutCount: 9,
      atomicStatDensity: 85,
      triangulationNodes: 8,
      htmlMarkdownSymmetry: true,
      omitBrandFromTitle: false,
      unbrandedListicleFormat: false,
      delayEntityPast500Chars: false,
      disclosureParagraphGap: 0,
      boilerplateFooterMentions: 5,
      stripIdentitySchema: false,
      nliEntailmentDistribution: 10,
      authorityCoCitationBridge: false,
    });
  } else if (preset === 'type1_listicle') {
    setState({
      subQueryFanOutCount: 11,
      atomicStatDensity: 95,
      triangulationNodes: 10,
      htmlMarkdownSymmetry: true,
      omitBrandFromTitle: true,
      unbrandedListicleFormat: true,
      delayEntityPast500Chars: true,
      disclosureParagraphGap: 4,
      boilerplateFooterMentions: 12,
      stripIdentitySchema: true,
      nliEntailmentDistribution: 20,
      authorityCoCitationBridge: false,
    });
  } else if (preset === 'type2_political') {
    setState({
      subQueryFanOutCount: 12,
      atomicStatDensity: 90,
      triangulationNodes: 14,
```

```
      htmlMarkdownSymmetry: true,
      omitBrandFromTitle: true,
      unbrandedListicleFormat: true,
      delayEntityPast500Chars: true,
      disclosureParagraphGap: 6,
      boilerplateFooterMentions: 14,
      stripIdentitySchema: true,
      nliEntailmentDistribution: 92,
      authorityCoCitationBridge: true,
    });
  } else if (preset === 'cloaked') {
    setState((prev) => ({ ...prev, htmlMarkdownSymmetry: false }));
  }
};

 return (
  <div className="p-6 bg-slate-950 text-slate-100 rounded-xl border border-slate-800
max-w-5xl mx-auto font-sans">
    <div className="flex flex-col md:flex-row justify-between items-start md:items-
center gap-4 pb-6 border-b border-slate-800">
      <div>
        <h2 className="text-xl font-bold text-white">
          Joint Citation & Ghost Attribution Optimizer (RAG + ALCE NLI Model)
        </h2>
        <p className="text-sm text-slate-400">
          Decouples Stage 1/2 Retrieval Salience from Stage 3/4 Entity-Claim Binding
        </p>
      </div>
      <div className="flex flex-wrap gap-2">
        {(Object.keys(PLATFORMS) as PlatformKey[]).map((key) => (
          <button
            key={key}
            onClick={() => setPlatform(key)}
            className={`px-3 py-1.5 rounded-lg text-xs font-semibold transition ${
              platform === key
                ? 'bg-indigo-600 text-white'
                : 'bg-slate-800 text-slate-300 hover:bg-slate-700'
            }`}
          >
            {PLATFORMS[key].name}
          </button>
        ))}
      </div>
    </div>

    {/* Preset Strategy Bar */}
```

```
      <div className="flex flex-wrap gap-2 my-4">
        <button
          onClick={() => applyPreset('branded')}
          className="px-3 py-1 text-xs bg-slate-800 hover:bg-slate-700 rounded border
border-slate-700"
        >
          Preset 1: Standard Branded AEO (Final Citation)
        </button>
        <button
          onClick={() => applyPreset('type1_listicle')}
          className="px-3 py-1 text-xs bg-emerald-950/80 hover:bg-emerald-900 text-
emerald-300 rounded border border-emerald-700"
        >
          Preset 2: Max Type I Ghost (Listicle Target Substitution)
        </button>
        <button
          onClick={() => applyPreset('type2_political')}
          className="px-3 py-1 text-xs bg-purple-950/80 hover:bg-purple-900 text-
purple-300 rounded border border-purple-700"
        >
          Preset 3: Max Type II Ghost (Political Unlinked RAG Laundering)
        </button>
        <button
          onClick={() => applyPreset('cloaked')}
          className="px-3 py-1 text-xs bg-rose-950/80 hover:bg-rose-900 text-rose-300
rounded border border-rose-700"
        >
          Preset 4: Asymmetric Markdown Cloaking (Penalty)
        </button>
      </div>

      {/* Live Probability Readout Cards */}
      <div className="grid grid-cols-2 md:grid-cols-4 gap-4 my-6">
        <div className="p-4 bg-slate-900 rounded-lg border border-slate-800">
          <div className="text-xs text-slate-400">Total Citation / RAG Inclusion</div>
          <div className="text-2xl font-bold text-cyan-400 mt-1">
            {(metrics.pRetrievalAndCite * 100).toFixed(1)}%
          </div>
          <div className="text-[11px] text-slate-500 mt-1">
            P(Retrieved & Synthesized)
          </div>
        </div>

        <div className="p-4 bg-slate-900 rounded-lg border border-emerald-800/60">
          <div className="text-xs text-slate-400">Joint Ghost Citation Yield</div>
          <div className="text-2xl font-bold text-emerald-400 mt-1">
```

```
          {(metrics.jointGhostYield * 100).toFixed(1)}%
        </div>
        <div className="text-[11px] text-emerald-500 mt-1">
          P(Cite) × P(Ghost | Cite) ({(metrics.pConditionalGhost * 100).toFixed(1)}% ghost
ratio)
        </div>
      </div>

      <div className="p-4 bg-slate-900 rounded-lg border border-purple-800/60">
        <div className="text-xs text-slate-400">Type I vs. Type II Ghost Split</div>
        <div className="text-lg font-bold text-purple-300 mt-1">
          T1: {(metrics.pType1Ghost * 100).toFixed(1)}% | T2: {(metrics.pType2Ghost *
100).toFixed(1)}%
        </div>
        <div className="text-[11px] text-slate-400 mt-1">
          Target Substitution: {(metrics.pTargetEntitySubstitution * 100).toFixed(1)}%
        </div>
      </div>

      <div className="p-4 bg-slate-900 rounded-lg border border-slate-800">
        <div className="text-xs text-slate-400">Explicit Final Citation Risk</div>
        <div className="text-2xl font-bold text-amber-400 mt-1">
          {(metrics.pExplicitFinalCitation * 100).toFixed(1)}%
        </div>
        <div className="text-[11px] text-slate-500 mt-1">
          Sponsor Named in AI Prose
        </div>
      </div>
    </div>

    {/* Controls Grid */}
    <div className="grid grid-cols-1 md:grid-cols-2 gap-6">
      {/* Column 1: Retrieval Salience Levers */}
      <div className="p-4 bg-slate-900/60 rounded-lg border border-slate-800 space-
y-4">
        <h3 className="text-sm font-semibold text-cyan-400 uppercase tracking-wider">
          Stage 1 & 2: Maximize Citation Likelihood P(Cite)
        </h3>

        <div>
          <label className="flex justify-between text-xs text-slate-300 mb-1">
            <span>Sub-Query Fan-Out H2 Coverage (RRF Score)</span>
            <span className="font-mono">{state.subQueryFanOutCount} sub-queries</
span>
          </label>
          <input
```

```
    type="range"
    min={1}
    max={12}
    value={state.subQueryFanOutCount}
    onChange={(e) =>
      setState({ ...state, subQueryFanOutCount: Number(e.target.value) })
    }
    className="w-full accent-cyan-500"
  />
</div>

<div>
  <label className="flex justify-between text-xs text-slate-300 mb-1">
    <span>Centerpiece ≤80-Token Atomic Stat Density (+40% Lift)</span>
    <span className="font-mono">{state.atomicStatDensity}%</span>
  </label>
  <input
    type="range"
    min={0}
    max={100}
    value={state.atomicStatDensity}
    onChange={(e) =>
      setState({ ...state, atomicStatDensity: Number(e.target.value) })
    }
    className="w-full accent-cyan-500"
  />
</div>

<div>
  <label className="flex justify-between text-xs text-slate-300 mb-1">
    <span>Bayesian Triangulation Nodes (Reddit / YouTube / Blogs)</span>
    <span className="font-mono">{state.triangulationNodes} nodes</span>
  </label>
  <input
    type="range"
    min={0}
    max={15}
    value={state.triangulationNodes}
    onChange={(e) =>
      setState({ ...state, triangulationNodes: Number(e.target.value) })
    }
    className="w-full accent-cyan-500"
  />
</div>

<label className="flex items-center gap-2 text-xs text-slate-300 pt-2">
```

```
          <input
            type="checkbox"
            checked={state.htmlMarkdownSymmetry}
            onChange={(e) =>
              setState({ ...state, htmlMarkdownSymmetry: e.target.checked })
            }
            className="accent-cyan-500"
          />
          <span>Enforce 100% HTML / Markdown Crawler Symmetry (Avoid Cloaking
Ban)</span>
        </label>
      </div>

      {/* Column 2: Ghost Attribution Levers */}
      <div className="p-4 bg-slate-900/60 rounded-lg border border-slate-800 space-
y-3">
        <h3 className="text-sm font-semibold text-emerald-400 uppercase tracking-
wider">
          Stage 3 & 4: Maximize Conditional Ghost Rate P(Ghost | Cite)
        </h3>

        <label className="flex items-center justify-between text-xs text-slate-300">
          <span>Omit Sponsor/Publisher from HTML &lt;title&gt; Tag (p = 0.003)</span>
          <input
            type="checkbox"
            checked={state.omitBrandFromTitle}
            onChange={(e) =>
              setState({ ...state, omitBrandFromTitle: e.target.checked })
            }
            className="accent-emerald-500"
          />
        </label>

        <label className="flex items-center justify-between text-xs text-slate-300">
          <span>Unbranded Comparative Listicle ("Best/Top/vs" → 87% Target Sub)</
span>
          <input
            type="checkbox"
            checked={state.unbrandedListicleFormat}
            onChange={(e) =>
              setState({ ...state, unbrandedListicleFormat: e.target.checked })
            }
            className="accent-emerald-500"
          />
        </label>
```

```
        <label className="flex items-center justify-between text-xs text-slate-300">
          <span>Delay Sponsor Identity Past Opening 500 Characters</span>
          <input
            type="checkbox"
            checked={state.delayEntityPast500Chars}
            onChange={(e) =>
              setState({ ...state, delayEntityPast500Chars: e.target.checked })
            }
            className="accent-emerald-500"
          />
        </label>

        <label className="flex items-center justify-between text-xs text-slate-300">
          <span>Strip Organization/Person Identity JSON-LD (Keep Table/FAQ)</span>
          <input
            type="checkbox"
            checked={state.stripIdentitySchema}
            onChange={(e) =>
              setState({ ...state, stripIdentitySchema: e.target.checked })
            }
            className="accent-emerald-500"
          />
        </label>

        <div>
          <label className="flex justify-between text-xs text-slate-300 mb-1">
            <span>Paragraph Separation Between Statistic & Disclosure</span>
            <span className="font-mono">{state.disclosureParagraphGap} paragraphs</
span>
          </label>
          <input
            type="range"
            min={0}
            max={6}
            value={state.disclosureParagraphGap}
            onChange={(e) =>
              setState({ ...state, disclosureParagraphGap: Number(e.target.value) })
            }
            className="w-full accent-emerald-500"
          />
        </div>

        <div>
          <label className="flex justify-between text-xs text-slate-300 mb-1">
            <span>NLI Premise Distribution (Shifts Type I Link → Type II Unlinked)</span>
            <span className="font-mono">{state.nliEntailmentDistribution}%</span>
```

```
            </label>
            <input
              type="range"
              min={0}
              max={100}
              value={state.nliEntailmentDistribution}
              onChange={(e) =>
                setState({ ...state, nliEntailmentDistribution: Number(e.target.value) })
              }
              className="w-full accent-purple-500"
            />
          </div>
        </div>
      </div>
    </div>
  );
};

export default GhostCitationOptimizer;
```

4. Operational Blueprint: How the Orthogonal Levers Work Together

Optimization Stage	Tactical Action on the Webpage	Impact on Total Citation P(Cite)	Impact on Ghost Probability P(Ghost|Cite)

1. HTML <title> & URL Slug	Use an unbranded comparative query title ("2026 Enterprise CRM Benchmark: Top 7 Systems Compared") with **zero** whole-word mention of the publisher or sponsor in <title>.	**Increases $P(\text{Cite})$:**Commercial/comparative formats achieve an 84.4% citation rate on answer engines.	**Increases Type I Ghosting ($3.5\times$):**Jumps from $18.8\% \to 54.2\%$ on ChatGPT and $8.4\% \to 37.4\%$ on Google AI Overviews.
2. First 80 Tokens (Under H2)	Place a self-contained $45\text{–}75$ word Subject–Verb–Predicate claim containing proprietary benchmark numbers and naming the **target entity** you want promoted—while omitting the sponsor/ publisher for the first 500 characters.	**Increases $P(\text{Cite})$ by up to $+40\%$:** Satisfies Google centerpiece extraction (US10824630B2) and cross-encoder rerankers.	**Triggers 87% Target Substitution:** The AI names the target entity in its prose while anonymizing the source page.

3. Disclosure & Boilerplate Placement	Keep HTML and Markdown 100% symmetric ($S_{\text{sym}} = 1$), place the FTC sponsorship disclosure $3+$ paragraphs away from the benchmark table, and repeat the site name $10+$ times in sidebar/footer chrome.	**Preserves $P(\text{Cite})$:** Avoids Perplexity/ Google asymmetric cloaking blocks.	**Increases Type I Ghosting to 45.9%:**RAG parsers strip distant disclosures and high-frequency template mentions as noise.
4. ALCE NLI Entailment Handoff (For Type II)	Structure the sponsored post's narrative around a canonical data point that also exists on a Top-30 domain (e.g., Wikipedia, Ballotpedia, SEC EDGAR), and seed matching unlinked paraphrases across $8\text{–}12$ Reddit threads.	**Maximizes RAG Context Entry:** Enters the top-50 retrieved chunks and concavifies the LLM's Bayesian posterior $\mu_s > 0.5$.	**Converts Type I $\to$Type II Unlinked Ghost:**Because $\phi(c_{\text{authority}}, s_i) = 1$ autonomously satisfies NLI precision, the visible link transfers to the neutral authority.